

MCP Servers & Custom Tool Integration

Agentic Workflows | Advanced

Agenda

Time	Topic
0:00–0:16	MCP model, discovery, and transports
0:16–0:26	Local configuration
0:26–0:34	Approved local demo
0:34–0:48	Custom server walkthrough
0:48–1:00	Safety review and lab handoff

MCP model

Part	Meaning
Client	AI surface that requests work
Server	Process or service exposing capabilities
Transport	stdio, SSE, or Streamable HTTP
Tool	Callable capability
Resource	Data available from the server

The client discovers tools, calls one, and returns the result to the model.

Local configuration

```
{
  "servers": {
    "sqlite": {
      "command": "uvx",
      "args": ["mcp-server-sqlite", "--db-path", "./data/app.db"]
    }
  }
}
```

Use `.vscode/mcp.json` for repository-shared configuration. Verify current setting names and supported behavior in official documentation.

A custom typed tool

```
server.tool(  
  "search_team",  
  "Search team members by name, role, or team. Returns matching results.",  
  { query: z.string().describe("Name, role, or team to search") },  
  async ({ query }) => ({ content: [{ type: "text", text: query }] })  
);
```

Clear descriptions and typed inputs help the model call the tool correctly.

Safety boundary

- Use supplied synthetic data only.
- Inspect the server source, tool descriptions, inputs, and outputs.
- Stop at the customer-defined metered-work guard.
- Do not connect credentials or customer systems.
- Session 17 covers organization policy, approval, registry, and rollout.

Lab

Build a weather server with `get_weather`, `get_forecast`, and `convert_temperature`. Test it locally, then use the supplied scenarios.